



'Featuristic' Linux Load-Balancing

This article deals with setting up LVS (Linux Virtual Server) for clustering on SuSE. Also discover how it acts as a scalable and high-performance delivery platform—just like an ideal load balancer.

Load balancing is a frequently used term in today's software industry. Any organisation that deals with Web services of any kind has to use load balancing to handle its Web traffic. Let me explain load balancing using the example of the biggest Web presence, Google. Let's look at the most basic Google query. Every second, across the globe, around a million queries are made to Google's search engine. To process these queries in milliseconds, you need an architecture that seldom fails, and scales and balances load very proficiently. Google has around a million servers around the globe. In this complicated architecture, there lies a configuration of load balancers of different platforms, out of which Linux load balancers are an increasing percentage, as they are better performers than the others.

When you enter a Google query, such as 'LFY', your browser first performs a domain name system (DNS) lookup to map www.google.com to a particular IP address. Load

balancing occurs over clusters, and a single cluster consists of a thousand machines. A DNS-based load-balancing system selects a cluster, by accounting for your geographic proximity to each physical cluster. The load-balancing system minimises the query processing time for your request, while also considering the available capacity at the various clusters. A hardware-based load balancer in each cluster monitors the available set of Google Web servers, and performs local load balancing of requests across a set of them. After receiving a query, a Web server machine coordinates the query execution, and formats the results into an HTML response to your browser. This process is depicted in Figure 1.

Load balancing can be categorised into three de-facto types:

1. *Load balancing through NAT:* This can be a bottleneck when it comes to a large number of clusters, because both requests and response have to be routed through the same load-balancer configuration.